

TINA REZVANIAN

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SUMMARY

Data science and ML expert with 7+ years driving innovative, user-facing products like Adobe Firefly, Liquid Mode, and Scan. Skilled in scalable machine learning, distributed systems, experimental design, causal analysis, and marketing optimization. Proficient in generative AI, LLMs, multimodal deep learning, and graph analytics, with expertise in predictive modeling, statistical analysis, and building scalable data pipelines. Successfully deployed 10+ models, enhancing technology integration, revenue, and efficiency.

EDUCATION

Northeastern University Ph.D., Industrial Engineering and Operations Research Minor in Mathematics	Boston, MA 2013 - 2019
Northeastern Illinois University MBA, Master of Business Administration	Chicago, IL 2011 - 2013
University of Science and Technology of Mazandaran B.Sc., Industrial Systems Engineering	Mazandaran, Iran 2006 - 2010

SKILLS

AI/ML/DS: Vision and Language Architectures (Transformers, CLIP), Diffusion Models, Attention Mechanisms, VAEs, Temporal CNNs, Reinforcement Learning, Causal Models, Supervised Unsupervised Learning, Ensemble Methods, SVMs, Dimensionality Reduction, Clustering, Auto-Labeling, Graph Models, Community Detection, Normalization Regularization, Optimization, XGBoost, GBM.

Programming Python, PyTorch, scikit-learn, Spacy, SQL, Git, Hadoop, Spark, TensorFlow, Docker, Kubernetes, RunAI, CI/CD Pipelines, R, MATLAB, Flask, AWS, CPLEX, Gurobi, SQL, Hadoop.

Visualization Matplotlib, Plotly, Seaborn, ggplot2, Gephi, Tableau, Power BI.

WORK EXPERIENCE

Adobe <i>Machine Learning Engineer, Data Scientist, Firefly</i>	Sep 2019 - May 2024 <i>San Jose, CA</i>
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- Firefly: Developed and finetuned multiple Firefly's modules such as input/output data encoder, input dense vector retrieval, diffusion Unet, and in-painting inference
 - . Enriched data with generated captions, GenAI images, and multi-representational embeddings.
 - . Developed evaluation models to assess coherence and technical quality of images using human-scores
- AB Testing Firefly: Evaluated the impact of data enrichment strategies and architectural modifications to address model limitations and enhance performance.
 - . Developed a rapid experimentation framework automating training and evaluation; integrating RunAI, AWS, Artifactory; streamlining job configurations, checkpoints, Docker, GPU, and data transfer.
- Gap Analysis: Identified subjects, contexts, and actions where Gen AI performance was suboptimal.
 - . Resolved data issues in text presence, aesthetic scoring, face/human detection, and captioning.
 - . Extracted insights from unsupervised similarity, categorization, caption alignment, and segmentation.
 - . Optimized caption weights by NLP analysis, focusing on alignment, part of speech, and model biases.
- Multi-Channel Attention Attribution: Boosted paid user retention and trial conversion by recommending next-best action through estimating the credit share of each touchpoint and its time-of-use.

- . Trained a scalable multi-head attention model with billions of parameters, visualized in a web app.
- . Identified critical workflows for engagement during the user journey
- Super Users: Segmented and characterized users to recommend workflow sequences, optimizing retention and personalizing onboarding, in-app, and email communications.
 - . Trained a β -variational auto-encoder to automatically extract meaningful attributes
 - . Generated similar sequences to that of retained users using the decoder within attribute-based clusters
- FontMART: Ranked fonts to maximize reading speed using LambdaMART, directly optimizing NDCG.
 - . Gathered and evaluated data from MTurk, web apps, and Hadoop to ensure quality and readiness.
 - . Identified optimal fonts, fastest speeds, and key font traits by age group.
- RealTime Recommender: Developed an on-device real-time recommender using reinforcement learning, trained via interactions with a generative AI user agent.
 - . Trained a generative adversarial network to model generalized user behavior from sequential data.
 - . Built a general deep Q-learning recommender and personalized it through user-specific fine-tuning.
- CausalTIM: Performed causal analysis to identify features' impact on conversion using observational data.
 - . Developed a parallel workflow for propensity score matching across features, handling data preparation, instance matching, and treatment effect estimation.

Harvard Extension School

Nov 2019 - May 2020

- Face Recognition: Developed face recognition models with CNNs to match extracted faces to identities in the CASIA-WebFace dataset.
 - . Adapted FaceNet, DeepFace, VGGFace, and ResNet models using feature extraction, transfer learning, fine-tuning, data augmentation, and regularization

Insight Data Science

Jan 2019 - Feb 2019

- PickaBasket: Developed a web application with Python, Flask, and AWS to offer bestseller bundles for online grocery stores using graph algorithms.
 - . Identified patterns in customers' purchase behavior as probabilistic rules on bestseller combinations.

Wheels UP Private Jet Company

May 2018 - Aug 2018

Data Science Intern

New York City, NY

- Forecasted daily flight demand and allocated aircraft to regions to minimize empty flights, clustering movements with regional metrics and community detection methods.

Northeastern University

Dec 2014 - Sep 2019

Research Assistant

Boston, MA

- Forecasted emergency response demand for refugees at UNHCR with a modified regression gradient boosting Tree (GBM), outperforming Tweedie and Poisson models for zero-inflated skewed variables.
- Developed a matching algorithm with negotiated Top Trading Cycle for multi-period markets on shortlists at the World Food Programme, delivering a Python-based decision support web app.
- Integrated emergency and ongoing supply chains in a multi-objective framework, reducing cost by 31% and response time by 18% with a two-stage stochastic model, delivering a Python-based decision support.

Healthcare Systems Engineering Institute (HSYE)

Sep 2013 - Dec 2014

Northeastern University, Research Assistant

Boston, MA

- Scheduled Treatments and Nurses using Column Generation Algorithm at Moffit Cancer Center.
- Optimized Curbside Threshold at Massachusetts General Hospital (MGH) Neurology Department.

PUBLICATIONS

- . Beier, Sofie, et al. "Readability research: An interdisciplinary approach." Foundations and Trends® in Human–Computer Interaction 16.4 (2022): 214-324.
- . Cai,T. , Wallace,S., **Rezvanian,T.**(2022, June). Personalized Font Recommendations:Combining ML and Typographic Guidelines to Optimize Readability. In Designing Interactive Systems Conference(pp.1-25)
- . Jahre,M.,Kembro,J., **Rezvanian,T.**, Ergun.O, Hapnes(2016)*Integrating supply chains for emergencies and ongoing operations in UNHCR*, Journal of Operations Management,45,57-72.
- . Jahre, M., Kembro, J., **Rezvanian, T.**, Ergun, O., Haapnes, S.,& Hultberg, M. *Scenario building in UNHCR-Predicting future demand in refugee crises*. In Proceedings of the 5th World Production and Operations Management Conference, POM Havana 2016
- . **Rezvanian, T.**, Ergun, O., Jahre, M., Kembro, J., (2020) *Forecasting Emergency Response Demand for Refugees as a Zero-Inflated Skewed variables: Evidence from UNHCR*
- . **Rezvanian, T.**, Ergun, O., *Two-sided Stable Matching with Negotiations*, Integrating Data-Driven Forecasting and Large-Scale Optimization to Improve Humanitarian Response Planning and Preparedness. Chapter 1, Diss. Northeastern University,
- . **Rezvanian, T.**, Ergun, O., *Deferred Acceptance Algorithms and Aftermarkets for Matching with Short Preference Lists at World Food Programme*, Integrating Data-Driven Forecasting and Large-Scale Optimization to Improve Humanitarian Response Planning and Preparedness. Chapter 1, Diss. Northeastern University, 2019.